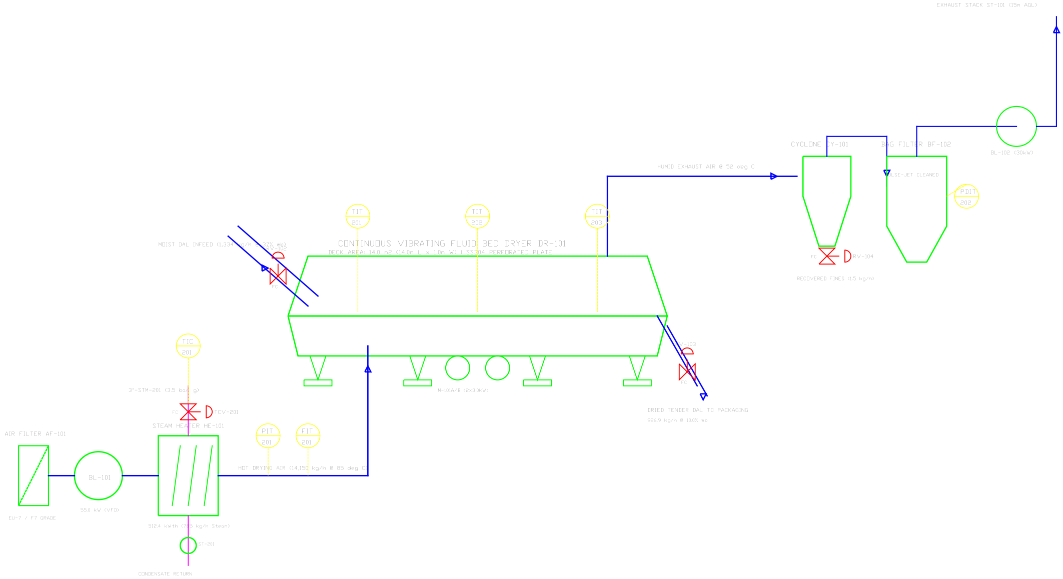
EXHAUST STACK ST-101 (15m AGL)
BL-102 (30kW)
CYCLONE CY-101
BAG FILTER BF-102
PULSE-JET CLEANED
PDIT
202
HUMID EXHAUST AIR @ 52 deg C
FC
RV-104
RECOVERED FINES (1.5 kg/h)
TIT
201
TIT
202
TIT
203
MOIST DAL INFEED (1,334 kg/h @ 37% wb)
RV-102
FC
CONTINUOUS VIBRATING FLUID BED DRYER DR-101
DECK AREA: 14.0 m2 (14.0m L x 1.0m W) | SS304 PERFORATED PLATE
M-101A/B (2x3.0kW)
RV-103
FC
DRIED TENDER DAL TO PACKAGING
926.9 kg/h @ 10.0% wb
TIC
201
3"-STM-201 (3.5 bar g)
FC
TCV-201
STEAM HEATER HE-101
AIR FILTER AF-101
BL-101
EU-7 / F7 GRADE
55.0 kW (VFD)
512.4 kWth (785 kg/h Steam)
ST-201
CONDENSATE RETURN
PIT
201
FIT
201
HOT DRYING AIR (14,150 kg/h @ 85 deg C)

DRYER CONTROL PHILOSOPHY & SAFETY INTERLOCK MATRIX (ISA-84 / SIL-1):

1. TIC-201: Modulates 3" steam valve TCV-201 to maintain fluidizing inlet air at 85 deg C +/- 1.5 deg C.
2. I-02 Interlock: Low airflow (FIT-201 < 10,000 kg/h) or Blower BL-101 trip immediately closes steam valve TCV-201.
3. I-03 Interlock: High-High bed temperature (TIT-202 > 95 deg C) trips infeed rotary valve RV-102 and alerts operator.
4. PDIT-202: Differential pressure across baghouse triggers sequential reverse pulse-jet cleaning @ 5.0 bar g air.
5. Vibration Interlock: High vibration switch VSH-101 trips exciter motors M-101A/B upon mechanical resonance.

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT

HARAKUTIR AGRO-PROCESS ENGINEERING PVT. LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

P&ID - FLUID BED DRYER & AIR SYSTEM
DISCIPLINE: PIPING | STAGE: CONCEPTUAL FEED

DWG NO: DWG-FIN-15T-005
REV: V4.0
SHEET: 5 OF 12
SCALE: AS SHOWN
DATE: 2026-10-15
STATUS: PRELIMINARY
GATE: G-A HOLD